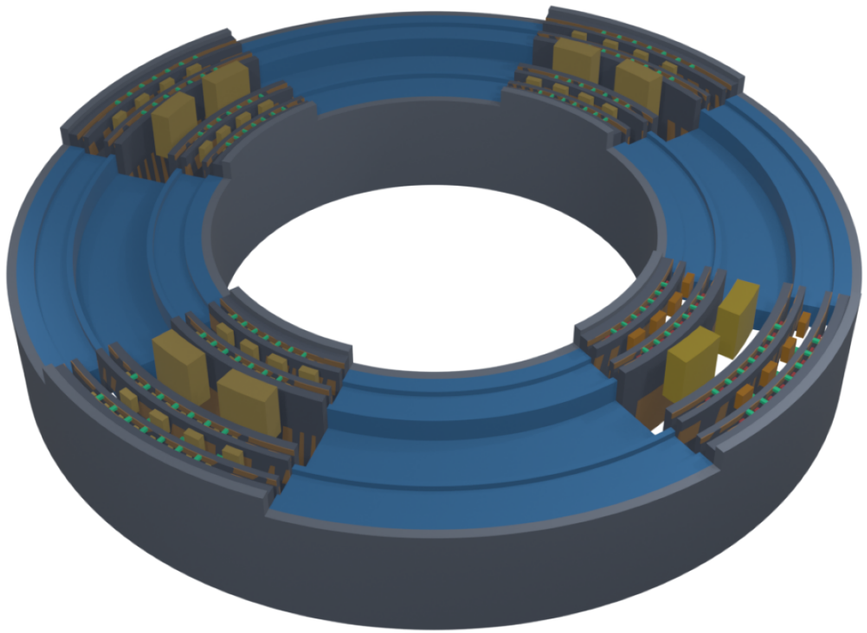
# H3-Cantor-Robust 3D Homothetic Architecture, Cleanroom Survival Features & Rotary-Axial Switching Kinematics

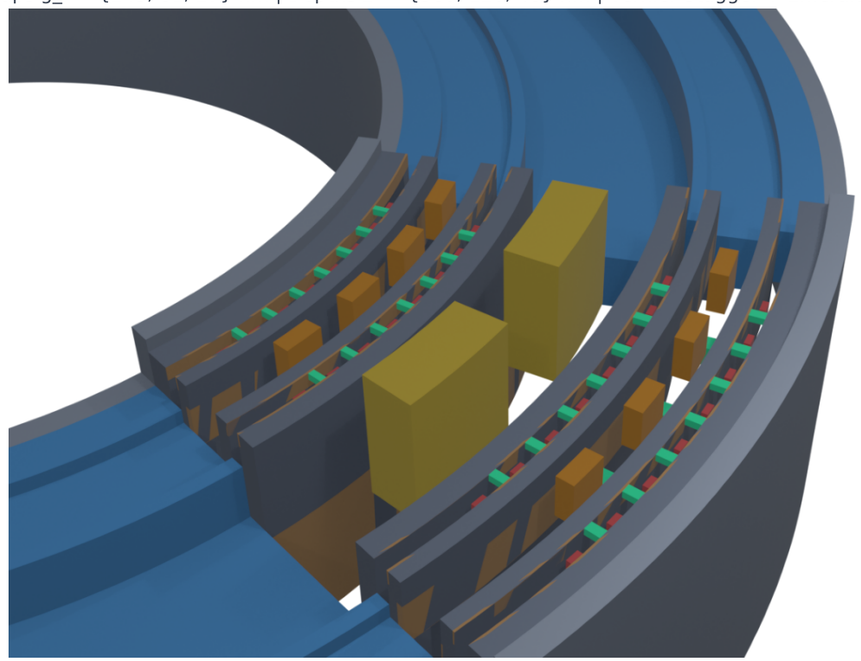
## (a) H3-Cantor-Robust Full C4 Rotary-Axial Switch (theta = 3 deg Disengaged)

R in [3.50, 6.20] um | A_tooth = 3.1026 um² | 7 Homothetic Tracks/Sector | Emerald Rim Micro-Bridges



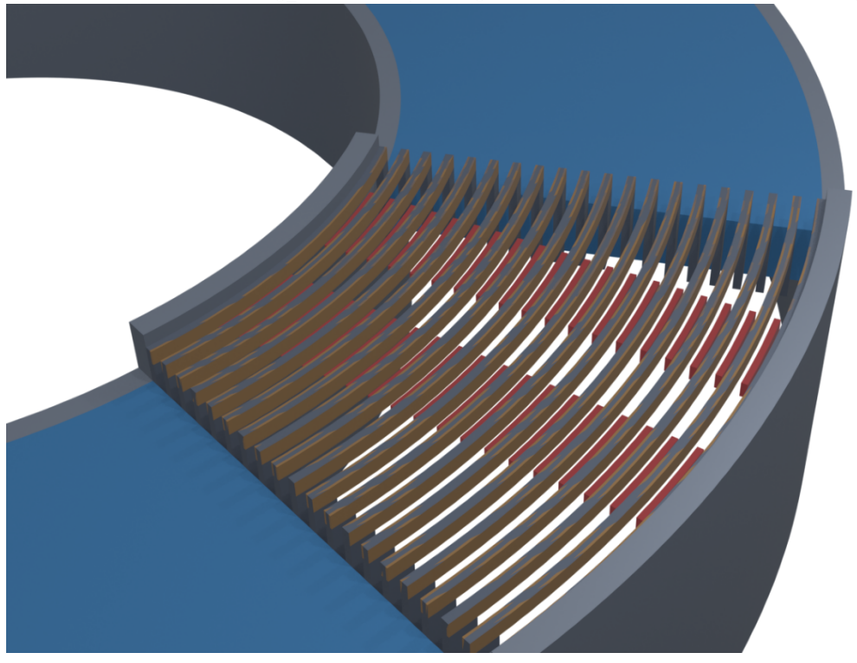
## (b) H3-Cantor-Robust Sector-0 Macro: Exact 9 : 3 : 1 (b = 3.0) 3D Homothety

Gaps g_r = {243, 81, 27} nm | Depths z = -{360, 120, 40} nm | Vernier Stagger + Emerald Rim Micro-Bridges



## (c) Ctrl-KnifeEdge-g27 Sector-0 Macro: Single-Scale 27 nm Point Optimizer

19 Coplanar Tracks (z = -40 nm, g_r = 27 nm) | 36 deg Unbridged 49.5 nm Stator Ribs -> Fragile Under Etch/Stroke



## (d) H3-Cantor-Robust Engaged Pad-Lock State (theta = 48 deg Attractive Latch)

Rotor Lobes Clear Emerald Rim Micro-Bridges & Latch Over Sapphire Stepped Pads (g_pad = {135, 45, 15} nm)

